



AI Research Group

CENTRE FOR ADVANCED STUDIES

Dr. A.P.J. Abdul Kalam Technical University
Lucknow



Research Seminar Series

Seminar -15

Deep learning based Sound Generation with different Approaches and Challenges

DATE & TIME:
07th September,
2022, 4:00 PM.

PLATFORM & WEB-LINK

Google Meet :
<https://meet.google.com/gjb-vwfj-yok>



SPEAKER
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ABSTRACT

Many researchers are abandoning the use of conventional signal processing techniques for sound generation as a result of the development of deep learning algorithms. Realistic sound textures, emotive voice synthesis, and musical notes from virtual instruments have all been accomplished by deep learning models. The ideal deep learning architecture, however, is currently being researched. The architecture choice and the audio representations are closely related. The original waveform of a sound may be too complicated and dense for deep learning models to handle effectively, and complexity lengthens training times and raises computational costs. Additionally, it doesn't accurately represent sound in the way that it is heard. Consequently, the raw audio has frequently been compressed and given a more comprehensible shape through feature extraction, upsampling, or even by using a higher level waveform illustration. In addition, several model architectures, extra conditioning representations, and metrics for measuring the reconstructed sound have all been researched conditional on the form used. The most important techniques for creating and assessing a sound synthesis architecture utilising deep learning models, always reliant on the audio representation, are also presented.